FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Non-Uniform Bodies

Questions by topic 10 questions · 115 marks

EXAM BOARDS Edexcel FM2 9FM0/4C

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*7 marks*	2
Question 2	*8 marks*	3
Question 3	*8 marks*	4
Question 4	*14 marks*	5
Question 5	*11 marks*	6
Question 6	*11 marks*	7
Question 7	*15 marks*	8
Question 8	*12 marks*	9
Question 9	*14 marks*	10
Question 10	*15 marks*	11

1. A thin rod AB of length $3a$ and mass M is non-uniform. At a distance x from A, its mass per unit length is

$$\rho = k\left(a^2 + x^2\right)$$

where k is a positive constant.

(a) Show that $k = \frac{M}{12a^3}$. **[3]**

(b) Find, in terms of a, the distance of the centre of mass of the rod from A. **[4]**

2. A non-uniform rod AB has length a. At a distance x from A, the mass per unit length of the rod is $k\left(1+e^{-x/a}\right)$, where k is a positive constant.
Show that the distance of the centre of mass of the rod from A is

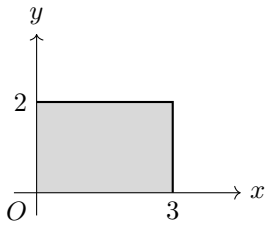
$$\frac{(3e-4)a}{2(2e-1)}$$

[8]

3. A non-uniform straight rod AB has length a. At a distance x from A, the mass per unit length of the rod is $kx\cos\left(\frac{\pi x}{2a}\right)$, where k is a positive constant and $0 \leq x \leq a$.

(a) Show that the mass of the rod is $\frac{2ka^2(\pi-2)}{\pi^2}$. **[3]**

(b) Find, in an exact form, the distance of the centre of mass of the rod from A. **[5]**

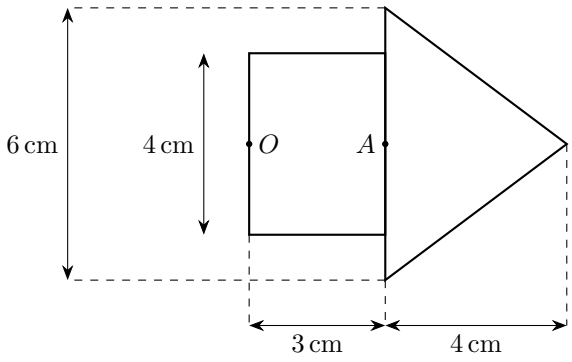


4. The shaded region shown in the diagram above is bounded by the x-axis, the line with equation $x = 3$, the line with equation $y = 2$ and the y-axis.
This region is rotated through 360° about the x-axis to form a solid of revolution. This solid is used to model a **non-uniform** cylinder of length 3 cm and radius 2 cm. The point O is the centre of the end of the cylinder in the plane $x = 0$, and the point A is the centre of the other end of the cylinder.
The mass per unit volume of the cylinder at the point (x, y, z) is $\lambda(12 - x^2)\,\text{kg cm}^{-3}$, where $0 \leq x \leq 3$ and λ is a positive constant.

(a) Show that the mass of the cylinder is $108\pi\lambda$ kg. **[3]**

(b) Show that the centre of mass of the cylinder is 1.25 cm from O. **[4]**

A uniform solid right circular cone has base radius 3 cm, height 4 cm and density $2\lambda\,\text{kg cm}^{-3}$. The cone is attached to the cylinder with the centre of its base in contact with the point A and its axis along the axis of the cylinder, to form the model shown in the diagram below.



The model is placed with the end containing O on a rough plane which is inclined at angle $\alpha°$ to the horizontal. The plane is sufficiently rough to prevent the model from sliding. The model is on the point of toppling.

(c) (i) Show that the centre of mass of the model is 1.75 cm from O. **[4]**

(ii) Find the value of α. **[3]**

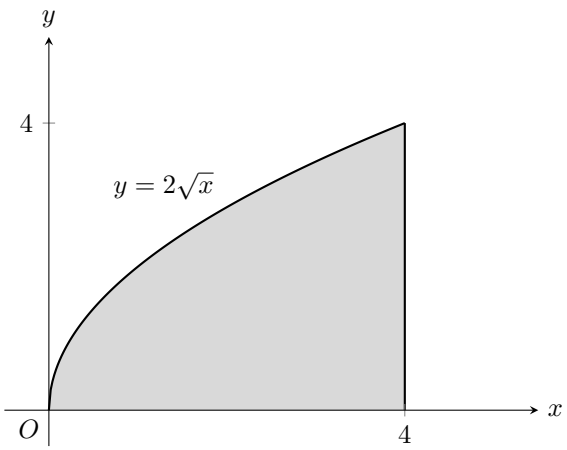
5. A solid right circular cone has height $2a$ m and base radius a m, and its vertex is V. At a point x m from the plane through V parallel to the base, the density of the cone, ρ kg m^{-3}, is given by

$$\rho = k\left(3 - \frac{x}{a}\right)$$

where k is a positive constant. The mass of the cone is M kg.

(a) Show by integration that $M = \pi k a^3$ and that the centre of mass of the cone is a distance $\frac{7}{5}a$ m from V. **[7]**

(b) A particle of mass m kg is attached to a point A on the circumference of the base of the cone. The cone is freely suspended from V and hangs in equilibrium with its axis making an angle θ with the vertical, where $\tan\theta = \frac{1}{4}$. Find m in terms of M. **[4]**

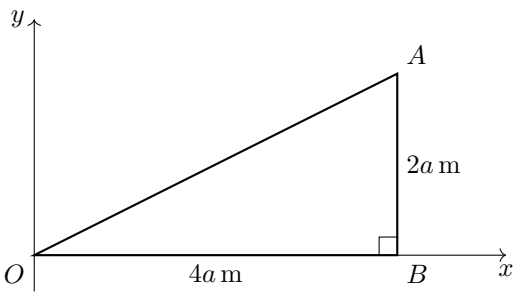


6. The shaded region shown in the diagram above is bounded by the x-axis, the line with equation $x = 4$ and the curve with equation $y = 2\sqrt{x}$, where x and y are measured in centimetres. This region is rotated through 360° about the x-axis to form a solid of revolution, which is used to model a solid paraboloid S with vertex O, height 4 cm and plane circular face of radius 4 cm.
The solid S is non-uniform and its mass per unit volume at the point (x, y, z) is $\lambda(6 - x)\,\text{kg cm}^{-3}$, where $0 \leq x \leq 4$ and λ is a positive constant.

(a) Find the distance of the centre of mass of S from O. **[6]**

A toy is made by joining the plane face of S to the plane face of a uniform solid hemisphere of radius 4 cm and density $\rho\,\text{kg cm}^{-3}$, so that the centres of the two plane faces coincide.
When the toy is placed on a smooth horizontal plane with any point of the curved surface of the hemisphere in contact with the plane, the toy remains at rest.

(b) Find λ in terms of ρ. **[5]**



7. A triangular lamina OAB of mass M kg has a right angle at B, with $OB = 4a$ m and $AB = 2a$ m. The diagram above shows the lamina in the x–y plane with origin O, with OB along the x-axis and BA parallel to the y-axis.
The mass per unit area, $\rho\,\text{kg m}^{-2}$, of the lamina at a point a distance y m from the x-axis is given by

$$\rho = k\left(1 + \frac{y}{a}\right)$$

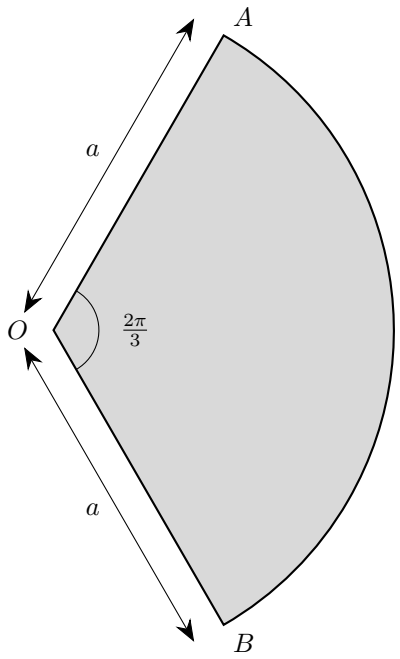
where k is a positive constant.

(a) By considering horizontal strips, show that $M = \frac{20}{3}ka^2$. **[5]**

(b) Use integration to find the coordinates of the centre of mass of the lamina, giving your answers in terms of a. **[7]**

(c) The lamina is freely suspended from A and hangs in equilibrium. Find the angle between AB and the vertical. **[3]**

8. *[In this question, you may assume that the centre of mass of a uniform circular arc of radius r, with angle at the centre 2α, is at a distance $\frac{r\sin\alpha}{\alpha}$ from the centre.]*



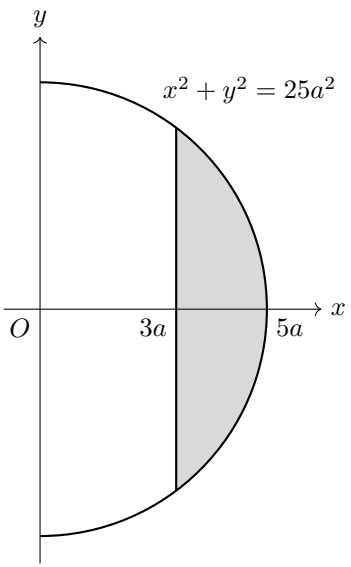
A thin non-uniform plate is in the shape of a sector OAB of a circle with centre O and radius a. The angle $AOB = \frac{2\pi}{3}$, as shown in the diagram above.
The plate is modelled as a non-uniform lamina. The mass per unit area of the lamina at any point P of the lamina is modelled as $k(2a - OP)$, where $k = \frac{3\lambda}{\pi a^3}$ and λ is a positive constant.
Using the model,

(a) find the mass of the plate in terms of λ, **[5]**

(b) show that the distance of the centre of mass of the plate from O is $\frac{15\sqrt{3}\,a}{16\pi}$. **[4]**

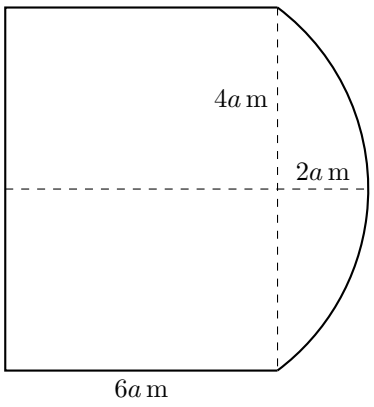
The plate is freely suspended from A and hangs in equilibrium.

(c) Find the angle between OA and the vertical, giving your answer in degrees to 1 decimal place. **[3]**



9. The shaded region shown in the diagram above is bounded by the line with equation $x = 3a$ and the curve with equation $x^2 + y^2 = 25a^2$.
This shaded region is rotated through 180° about the x-axis to form a solid of revolution.
This solid is used to model a dome with height $2a$ m and base radius $4a$ m.
The dome is modelled as being non-uniform, with the mass per unit volume of the dome at the point (x, y, z) equal to $\frac{\lambda}{x^2}$ kg m^{-3}, where $3a \leq x \leq 5a$ and λ is a positive constant.

(a) Show that the distance of the centre of mass of the dome from the centre of its plane face is $\left(\frac{75}{4} \ln \frac{5}{3} - 9\right) a$ m. **[6]**



A uniform solid right circular cylinder has radius $4a$ m and height $6a$ m. A toy is formed by joining the plane face of the dome to one plane face of the cylinder, as shown in the diagram above.
The weight of the cylinder is kW and the weight of the dome is $4W$.
The centre of mass of the toy is a distance d m from the plane face of the dome.

(b) Show that

$$d = \frac{\left|3k + 36 - 75 \ln \frac{5}{3}\right|}{k + 4} a$$ **[4]**

The toy is suspended from a point on the circumference of the plane face of the cylinder that is not joined to the dome, and hangs freely in equilibrium with its axis of symmetry at an angle β to the vertical.

(c) Given that $\tan \beta = \frac{4}{5}$, find the exact value of k. **[4]**

10. A non-uniform rod AB has length l and weight W. The weight per unit length of the rod at a distance x from A is proportional to $l + kx$, where k is a positive constant.

(a) Show that the centre of mass of the rod is at a distance $\frac{(3+2k)\,l}{3(2+k)}$ from A. **[6]**

In the case $k = 3$, the rod rests in a vertical plane with the end A on rough horizontal ground and with the rod in contact with a smooth horizontal rail at the point C of the rod, where $AC = \frac{4}{5}l$. The rail is perpendicular to the vertical plane containing the rod, and the rod makes an angle θ with the ground, where $0 < \theta < \frac{\pi}{2}$. The coefficient of friction between the rod and the ground is μ.

(b) Show that, for the rod to be in equilibrium,

$$\mu \geq \frac{3\tan\theta}{1 + 4\tan^2\theta}$$ **[7]**

(c) Hence find the least value of μ for which the rod can rest in equilibrium in this way whatever the value of θ. **[2]**